



Name: Azkiya Putri Ihtiarilah

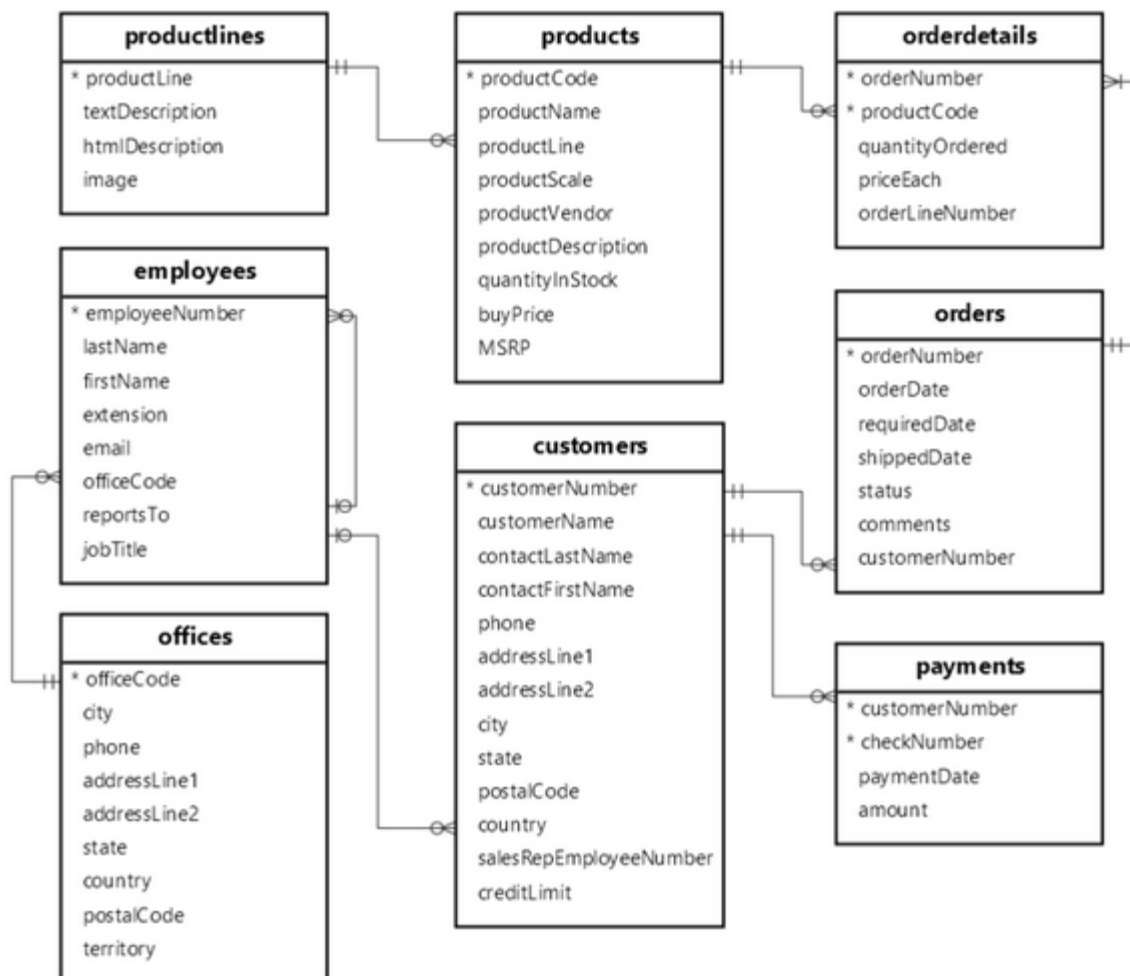
Nim: 2341760175

Tujuan Praktikum

Setelah melakukan praktikum ini, mahasiswa diharapkan dapat lebih mengenal data sumber, cara menganalisa serta melihat kebutuhan baik fungsional maupun non-fungsional dalam pengembangan data warehouse serta lebih memahami apa itu OLTP.

Studi Kasus

LegendVehicle merupakan perusahaan jual-beli tukar-tambah kendaraan klasik. Perusahaan ini memiliki cabang di berbagai negara. LegendVehicle memiliki sistem informasi ERP sendiri. Salah satu modul dari sistem ERP tersebut adalah modul penjualan. Desain database dari modul tersebut adalah sebagai berikut:



Selain itu proses penjualan kendaraan pada perusahaan tersebut bukan hanya melalui showroom cabang, melainkan reseller-reseller bebas lainnya.

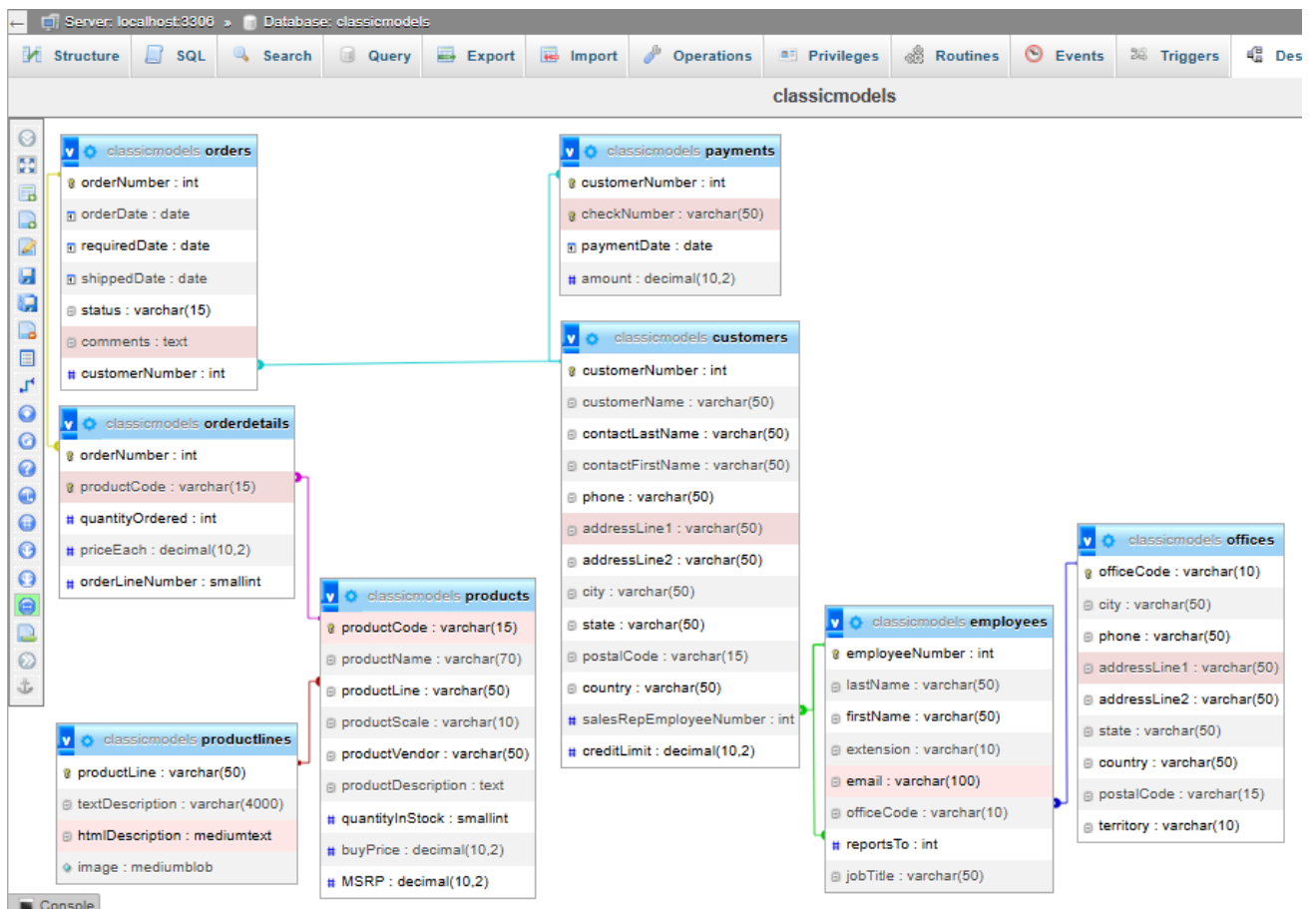
Data penjualan dari cabang tersebut dapat diunduh melalui link berikut:

 **mysqlsampledatabase.zip**
51.2 KB • Done

Tugas 1

1. Import data perusahaan tersebut pada DBMS MySQL!

Table	Action	Rows	Type	Collation	Size
☐ customers	★ Browse Structure Search Insert Empty Drop	122	InnoDB	utf8mb4_0900_ai_ci	64.0 KiB
☐ employees	★ Browse Structure Search Insert Empty Drop	23	InnoDB	utf8mb4_0900_ai_ci	48.0 KiB
☐ offices	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB
☐ orderdetails	★ Browse Structure Search Insert Empty Drop	2,996	InnoDB	utf8mb4_0900_ai_ci	32.0 KiB
☐ orders	★ Browse Structure Search Insert Empty Drop	326	InnoDB	utf8mb4_0900_ai_ci	64.0 KiB
☐ payments	★ Browse Structure Search Insert Empty Drop	273	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB
☐ productlines	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB
☐ products	★ Browse Structure Search Insert Empty Drop	110	InnoDB	utf8mb4_0900_ai_ci	80.0 KiB
8 tables	Sum	3,864	InnoDB	utf8mb4_0900_ai_ci	336.0 KiB



2. Analisa struktur data dari database perusahaan tersebut, dalam bentuk tabel, analisa hubungan setiap tabel nya!

Tabel 1	Tabel 2	Jenis Relasi
productlines	products	one to many
...	...	...
...	...	...

Tabel 1	Tabel 2	Tabel 3
productlines	product	1 : many
offices	employees	1 : many
employees	employees	1 : many
employees	customers	1: many

customers	payments	1 : many
customers	orders	1 : many
orders	ordersdetails	1 : many
products	orderdetails	1 : many

3. Analisa jumlah field pada setiap tabel!

Nama Tabel	Jumlah Field
...	...
...	...
...	...

Nama Tabel	Jumlah Fields
productlines	4
products	9
ordersdetails	5
offices	9
customers	13
payments	4
orders	7
employees	8

A. Analisa Data

PERINGATAN: jika menemukan "ERROR" maka, Beranilah untuk menemukan dimana letak kesalahan untuk memberikan solusi. Jangan hanya bisa menyalahkan namun, tidak dapat memberikan solusinya.

PRAKTIKUM 1

1. Jalankan **query** berikut pada **DBMS MySql** yang telah tersedia **data Perusahaan LegendVehicle**.

query
<pre>SELECT * FROM employees employe, employes manager, customer cust WHERE employee.reportsTo=manager.employeeNumber AND employee.employeeNumber=cust.salesRepEmployeeNumber;</pre>

maka hasil dari query tersebut adalah data Employee beserta Manajernya dan Customer yang ia miliki. perhatikan hasil data dengan seksama.

Error

SQL query: [Copy](#)

```
SELECT *
FROM employees employe, employes manager, customer cust
WHERE employee.reportsTo=manager.employeeNumber
AND employee.employeeNumber=cust.salesRepEmployeeNumber LIMIT 0, 25
```

MySQL said:

#1146 - Table 'classicmodels.employes' doesn't exist

Kesalahan terletak pada penulisan employees dan customers. Perbaiki kode:

```

1 SELECT *
2 FROM employees employee, employees manager, customers cust
3 WHERE employee.reportsTo=manager.employeeNumber
4 AND employee.employeeNumber=cust.salesRepEmployeeNumber;

```

Output Perbaikan Kode:

✓ Showing rows 0 - -1 (0 total, Query took 0.0012 seconds.)

```

SELECT * FROM employees employee, employees manager, customers cust WHERE
employee.reportsTo=manager.employeeNumber AND employee.employeeNumber=cust.salesRepEmployeeNumber;

```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

2. Buka **tab baru** pada browser untuk melakukan eksekusi **query** berikut:

query

```

SELECT manager.employeeNumber as id_manager,
CONCAT(manager.firstName," ",manager.lastName) as Manager,
employee.employeeNumber as id_staff,
CONCAT(employee.firstName," ",employee.lastName) as staff
FROM employees employee, employees manager
WHERE employee.reportsTo=manager.employeeNumber
ORDER BY manager.firstName;

```

dari hasil **query** diatas maka akan ditemukan atasan dari setiap pegawai.

Error

Static analysis:

1 errors were found during analysis.

1. Ending quote " was expected. (near "" at position 327)

SQL query: [Copy](#) [?](#)

```

SELECT manager.employeeNumber as id_manager, CONCAT(manager.firstName,"
,manager.lastName) as Manager, employee.employeeNumber as id_staff,
CONCAT(employee.firstName," ",employee.lastName) as staff FROM employees employee,
employees manager WHERE employee.reportsTo=manager.employeeNumber ORDER BY
manager.firstName;;

```

MySQL said: [?](#)

```

#1064 - You have an error in your SQL syntax; check the manual that corresponds
to your MySQL server version for the right syntax to use near
'",employee.lastName) as staff
FROM employees employee, employees manager
WHE' at line 4

```

Perbaikan kode:

✓ Showing rows 0 - 21 (22 total, Query took 0.0034 seconds.)

```

SELECT manager.employeeNumber as id_manager, CONCAT(manager.firstName,"
",manager.lastName) as Manager, employee.employeeNumber as id_staff,
CONCAT(employee.firstName," ",employee.lastName) as staff FROM employees
employee, employees manager WHERE employee.reportsTo=manager.employeeNumber
ORDER BY manager.firstName;

```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

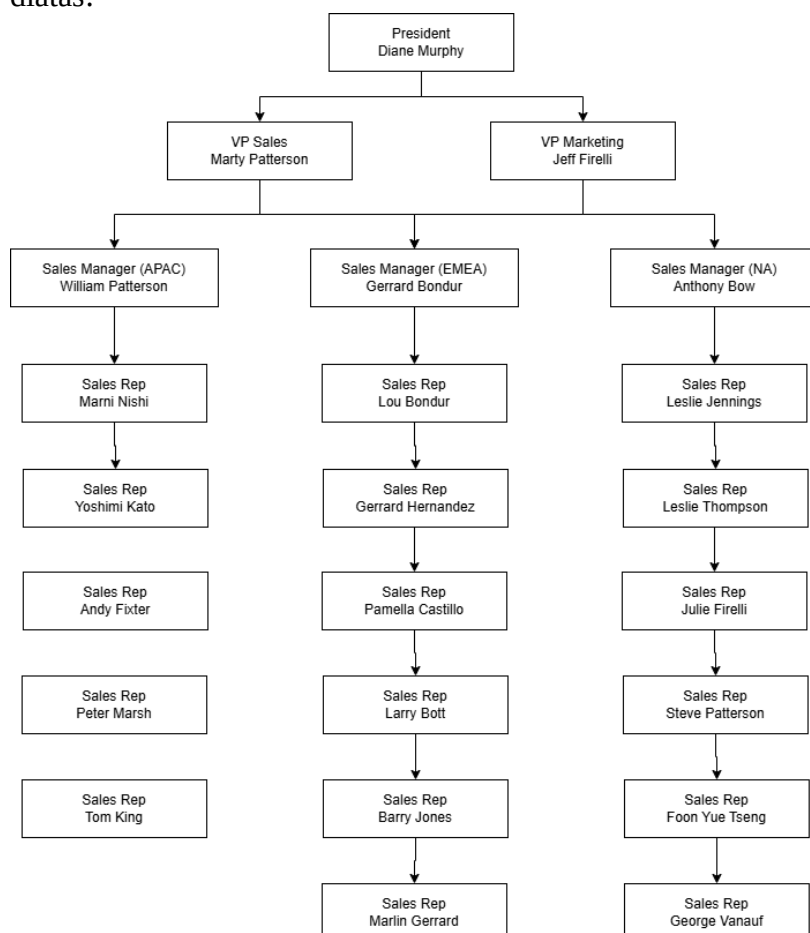
Output:

dari hasil **query** diatas maka akan ditemukan atasan dari setiap pegawai.

id_manager	Manager	id_staff	staff
1143	Anthony Bow	1165	Leslie Jennings
1143	Anthony Bow	1166	Leslie Thompson
1143	Anthony Bow	1188	Julie Firrelli
1143	Anthony Bow	1216	Steve Patterson
1143	Anthony Bow	1286	Foon Yue Tseng
1143	Anthony Bow	1323	George Vanauf
1002	Diane Murphy	1056	Mary Patterson
1002	Diane Murphy	1076	Jeff Firrelli
1102	Gerard Bondur	1337	Loui Bondur
1102	Gerard Bondur	1370	Gerard Hernandez
1102	Gerard Bondur	1401	Pamela Castillo
1102	Gerard Bondur	1501	Larry Bott
1102	Gerard Bondur	1504	Barry Jones
1102	Gerard Bondur	1702	Martin Gerard
1621	Mami Nishi	1625	Yoshimi Kato
1056	Mary Patterson	1088	William Patterson
1056	Mary Patterson	1102	Gerard Bondur
1056	Mary Patterson	1143	Anthony Bow
1056	Mary Patterson	1621	Mami Nishi
1088	William Patterson	1611	Andy Fixter
1088	William Patterson	1612	Peter Marsh
1088	William Patterson	1619	Tom King

PRAKTIKUM 2

1. Gambarkan hirarki organisasi berdasarkan atasan dari setiap pegawai sesuai dengan hasil prkatikum diatas!



2. Buka **tab baru** pada browser untuk melakukan eksekusi **query** berikut:

query

```
SELECT manager.employeeNumber as id_manager,  
concat(manager.firstName," ",manager.lastName) as Manager,  
employee.employeeNumber as id_staff, concat(employee.firstName," ",employee.lastName) as staff,  
count(cust.customerNumber) as total_cust  
FROM employees employee join employees manager on employee.reportsTomanager.employeeNumber  
left join customers cust on employee.employeeNumber=cust.salesRepEmployeeNumber  
GROUP BY employee.employeeNumber  
ORDER BY manager.firstName;
```

dari query tersebut menghasilkan jumlah **customer** dari setiap **staff**.

Error

SQL query: [Copy](#) 

```
SELECT manager.employeeNumber as id_manager,  
concat(manager.firstName," ",manager.lastName) as Manager,  
employee.employeeNumber as id_staff, concat(employee.firstName," ",employee.lastName) as staff,  
count(cust.customerNumber) as total_cust  
FROM employees employee join employees manager on employee.reportsTomanager.employeeNumber  
left join customers cust on employee.employeeNumber=cust.salesRepEmployeeNumber
```

MySQL said: 

#1054 - Unknown column 'employee.reportsTomanager.employeeNumber' in 'on clause'

Perbaiki kode:

Showing rows 0 - 21 (22 total, Query took 0.0037 seconds.)

```
SELECT manager.employeeNumber AS id_manager, CONCAT(manager.firstName, " ", manager.lastName) AS Manager,  
employee.employeeNumber AS id_staff, CONCAT(employee.firstName, " ", employee.lastName) AS staff,  
COUNT(cust.customerNumber) AS total_cust FROM employees employee JOIN employees manager ON  
employee.reportsTo = manager.employeeNumber LEFT JOIN customers cust ON employee.employeeNumber =  
cust.salesRepEmployeeNumber GROUP BY employee.employeeNumber ORDER BY manager.firstName;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

Output:

id_manager	Manager	id_staff	staff	total_cust
1143	Anthony Bow	1185	Leslie Jennings	6
1143	Anthony Bow	1186	Leslie Thompson	6
1143	Anthony Bow	1188	Julie Firrelli	6
1143	Anthony Bow	1216	Steve Patterson	6
1143	Anthony Bow	1286	Foon Yue Tseng	7
1143	Anthony Bow	1323	George Vanauf	8
1002	Diane Murphy	1056	Mary Patterson	0
1002	Diane Murphy	1076	Jeff Firrelli	0
1102	Gerard Bondur	1337	Loui Bondur	6
1102	Gerard Bondur	1370	Gerard Hernandez	7
1102	Gerard Bondur	1401	Pamela Castillo	10
1102	Gerard Bondur	1501	Larry Bott	8
1102	Gerard Bondur	1504	Barry Jones	9
1102	Gerard Bondur	1702	Martin Gerard	6
1621	Mami Nishi	1625	Yoshimi Kato	0
1056	Mary Patterson	1088	William Patterson	0
1056	Mary Patterson	1102	Gerard Bondur	0
1056	Mary Patterson	1143	Anthony Bow	0
1056	Mary Patterson	1621	Mami Nishi	5
1088	William Patterson	1611	Andy Fixter	5
1088	William Patterson	1612	Peter Marsh	5
1088	William Patterson	1619	Tom King	0

Jika perusahaan tersebut memiliki **KPI (Key Performances Indicator) "Jumlah customer yang bertransaksi"** maka jawablah pertanyaan-pertanyaan berikut!

PRAKTIKUM 3

- Siapaakah staff dengan hirarki paling bawah yang berprestasi dilihat dari jumlah customer terbanyak?
Staff dengan hirarki terbawah adalah **Pamela Castillo, employee number: 1401, with 10 total customers**
- Jika KPI atasan dihitung dari customer yang dimilikinya dijumlah dengan customer dari staff dibawahnya, urutkan ranking prestasi keseluruhan pegawai beserta keterangan jumlah customer yang dimilikinya!

Rank	Employee	Total Customer	Keterangan
1	Gerrand Bondur	46	6 bawahan (6+7+10+8+9+6)
2	Anthony Bow	33	6 bawahan (6+6+0+6+7+8)
3	William Patterson	15	5 bawahan (5+0+5+5+0)

- Analisa kembali data LegendVehicle untuk mendapatkan ranking pegawai berdasarkan KPI "**Jumlah omset yang didapat**". Urutkan ranking pegawai beserta keterangan dana yang didapat!

query

Showing rows 0 - 14 (15 total, Query took 0.0096 seconds.)

```
SELECT e.employeeNumber, CONCAT(e.firstName, " ", e.lastName) AS employee_name, SUM(od.quantityOrdered * od.priceEach) AS total_sales FROM employees e JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber JOIN orders o ON c.customerNumber = o.customerNumber JOIN orderdetails od ON o.orderNumber = od.orderNumber GROUP BY e.employeeNumber ORDER BY total_sales DESC;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Output:

employeeNumber	employee_name	total_sales
1370	Gerard Hernandez	1258577.81
1165	Leslie Jennings	1081530.54
1401	Pamela Castillo	868220.55
1501	Larry Bott	732096.79
1504	Barry Jones	704853.91
1323	George Vanauf	669377.05
1612	Peter Marsh	584593.76
1337	Loui Bondur	569485.75
1611	Andy Fixter	562582.59
1216	Steve Patterson	505875.42
1286	Foon Yue Tseng	488212.67
1621	Mami Nishi	457110.07
1702	Martin Gerard	387477.47
1188	Julie Firrelli	386663.20
1166	Leslie Thompson	347533.03

- Jika KPI yang pertama merupakan "**Jumlah customer yang bertransaksi**" sedangkan KPI yang kedua "**Jumlah omset yang didapat**". Maka, berapakah jumlah field yang dibutuhkan untuk mendapatkan informasi tersebut?

KPI 1 - **Jumlah customer yang bertransaksi**: 1 field, nomor pelanggan(customerNumber)

KPI 2 - **Jumlah omset yang didapat**:

3 field,

- barang yang dibeli(quantityOrdered)
- harga barang(priceEach)
- nomor pelanggan(customerNumber)

5. Buatlah report pertahun untuk KPI "**Jumlah omset yang didapat**" pada **Foon Yue Tseng** dan **Pamela Castillo**. Serta gambarkan grafiknya (grafik garis).

query

Showing rows 0 - 5 (6 total, Query took 0.0037 seconds.) [firstName: FOON YUE... - PAMELA...]

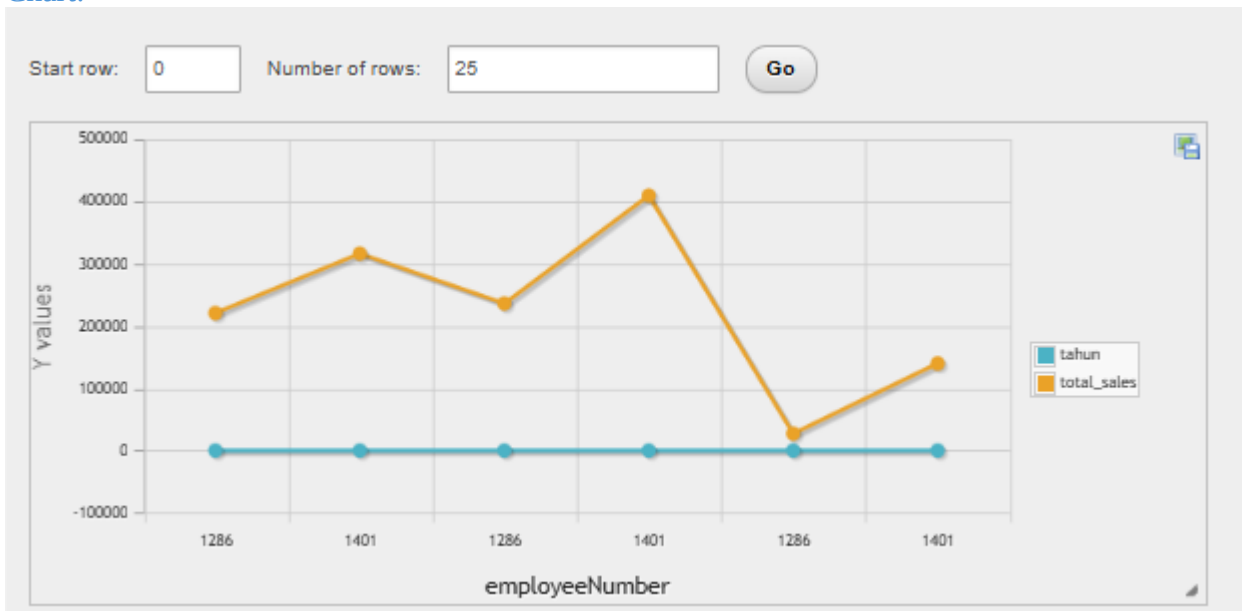
```
SELECT e.employeeNumber, e.firstName, e.lastName, YEAR(o.orderDate) AS tahun, SUM(od.quantityOrdered * od.priceEach) AS total_sales FROM employees e JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber JOIN orders o ON c.customerNumber = o.customerNumber JOIN orderdetails od ON o.orderNumber = od.orderNumber WHERE e.firstName IN ('Foon Yue', 'Pamela') AND e.lastName IN ('Tseng', 'Castillo') GROUP BY e.employeeNumber, tahun ORDER BY tahun, e.firstName;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Output:

employeeNumber	firstName	lastName	tahun	total_sales
1286	Foon Yue	Tseng	2003	221887.03
1401	Pamela	Castillo	2003	317104.78
1286	Foon Yue	Tseng	2004	237255.26
1401	Pamela	Castillo	2004	409910.07
1286	Foon Yue	Tseng	2005	29070.38
1401	Pamela	Castillo	2005	141205.70

Chart:



Penjelasan grafik

Performa Karyawan 1286 (Foon Yue Tseng)

Tahun 2003: Penjualan sebesar 221,887.03.

Tahun 2004: Penjualan meningkat menjadi 237,255.26.

Tahun 2005: Penjualan turun drastis menjadi 29,070.38.

Performa Karyawan 1401 (Pamela Castillo)

Tahun 2003: Penjualan sebesar 317,104.78.

Tahun 2004: Penjualan meningkat signifikan menjadi 409,910.07.

Tahun 2005: Penjualan turun menjadi 141,205.70.

Grafik ini menunjukkan pola "peak and drop," yaitu kinerja puncak di tengah periode (2004) diikuti oleh penurunan drastis di akhir periode (2005). Penurunan ini kemungkinan dipengaruhi oleh faktor eksternal seperti strategi perusahaan atau kondisi pasar.

Studi Kasus

Pak Huhut merupakan pemegang saham LegendVehicle. dia membutuhkan dashboard untuk melihat perkembangan penjualan (omset) di setiap cabang di tiap tahunnya. Dikarenakan perusahaan tersebut belum merekrut Data Engineer maka, penarikan informasi hanya bisa dilakukan melalui OLTP yang ada.

Hasil report yang diinginkan adalah grafik berdasarkan tabel berikut:

Nama Cabang	2003	2004	2005
...			
...			

Analisalah terlebih dahulu:

1. Field apa saja yang diperlukan untuk menampilkan penjualan di setiap cabang.

Field	Keterangan
branchName	Nama cabang
orderDate	Tanggal Waktu transaksi
orderAmount	Nilai total transaksi
branchID	Id tiap cabang

2. Bentuk query dengan memperhatikan relasi antar tabel.

query
✔ Showing rows 0 - 20 (21 total, Query took 0.0058 seconds.) <pre>SELECT o.city AS branchName, YEAR(p.paymentDate) AS Tahun, SUM(p.amount) AS Total_Omset FROM payments p JOIN customers c ON p.customerNumber = c.customerNumber JOIN employees e ON c.salesRepEmployeeNumber = e.employeeNumber JOIN offices o ON e.officeCode = o.officeCode GROUP BY o.city, YEAR(p.paymentDate) ORDER BY o.city, YEAR(p.paymentDate);</pre> ☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Output:

branchName	Tahun	Total_Omset
Boston	2003	301781.38
Boston	2004	467177.07
Boston	2005	66923.88
London	2003	505384.85
London	2004	674815.75
London	2005	144125.30
NYC	2003	391175.53
NYC	2004	623872.78
NYC	2005	57571.16
Paris	2003	969959.90
Paris	2004	1368458.96
Paris	2005	480750.04
San Francisco	2003	532681.13
San Francisco	2004	517408.62
San Francisco	2005	287349.83
Sydney	2003	281985.51
Sydney	2004	509833.62
Sydney	2005	215473.85
Tokyo	2003	267249.40
Tokyo	2004	151761.45
Tokyo	2005	38099.22

SOAL BONUS: buatlah report lain dengan sumber data OLTP yang sama, analisa field yang digunakan, bentuk struktur query dan tuliskan dalam tabel serta grafiknya.

Report Performa Produk terlaris Cabang per Tahun

Field	Table	Kegunaan
officeCode	Offices	Identitas cabang
city	Offices	Nama cabang
employeeNumber	employees	Karyawan penanggung jawab customer
SalesRepEmployeeNumber	customers	Karyawan yang handle customer
customerNumber	customers	Identitas customer
orderNumber	orders	Kunci transaksi
Orderdate	orders	Tanggal Transaksi
Productcode	orderdetails	Produk yang dipesan
Quantityordered	orderdetails	Berapa unit produk yang dipesan
Productname	products	Nama produk
productline	products	Kategori produk

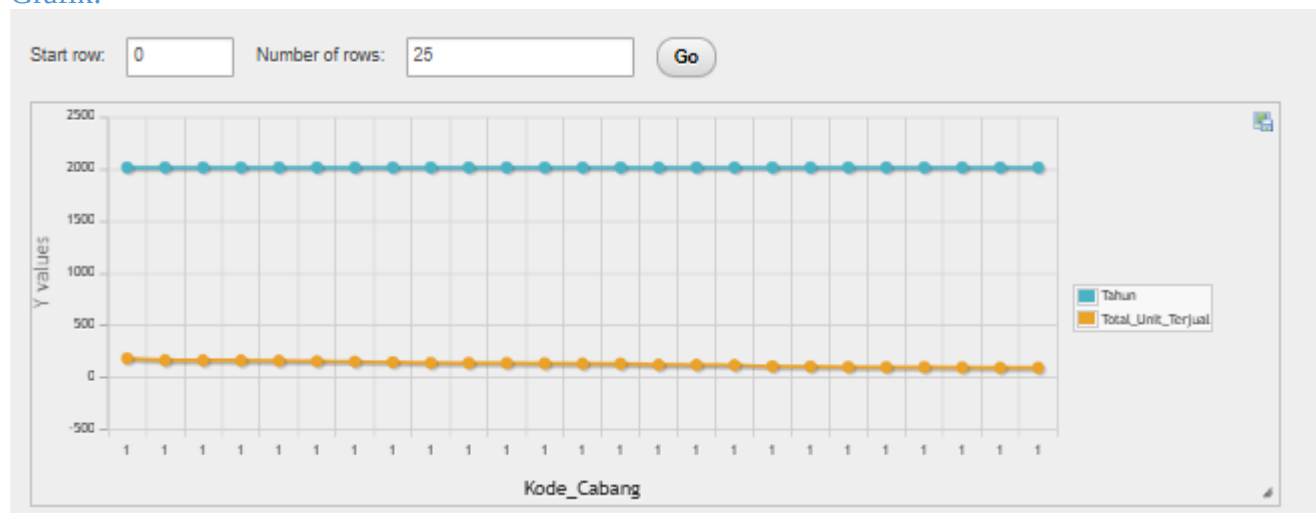
membentuk struktur query:

query
 Showing rows 0 - 24 (1539 total, Query took 0.0382 seconds.) <pre> SELECT o.officeCode AS Kode_Cabang, o.city AS Nama_Cabang, YEAR(od.orderDate) AS Tahun, p.productName AS Produk, SUM(odd.quantityOrdered) AS Total_Unit_Terjual FROM offices o JOIN employees e ON o.officeCode = e.officeCode JOIN customers c ON e.employeeNumber = c.salesRepEmployeeNumber JOIN orders od ON c.customerNumber = od.customerNumber JOIN orderdetails odd ON od.orderNumber = odd.orderNumber JOIN products p ON odd.productCode = p.productCode GROUP BY o.officeCode, o.city, Tahun, Produk ORDER BY o.officeCode, Tahun, Total_Unit_Terjual DESC; </pre> ☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Output:

Kode_Cabang	Nama_Cabang	Tahun	Produk	Total_Unit_Terjual
1	San Francisco	2003	1997 BMW R 1100 S	176
1	San Francisco	2003	1939 Chevrolet Deluxe Coupe	157
1	San Francisco	2003	1937 Lincoln Berline	156
1	San Francisco	2003	1936 Mercedes-Benz 500K Special Roadster	155
1	San Francisco	2003	1969 Harley Davidson Ultimate Chopper	152
1	San Francisco	2003	1936 Harley Davidson El Knucklehead	148
1	San Francisco	2003	1938 Cadillac V-16 Presidential Limousine	145
1	San Francisco	2003	2002 Suzuki XREO	140
1	San Francisco	2003	1932 Model A Ford J-Coupe	133
1	San Francisco	2003	1936 Mercedes Benz 500k Roadster	132
1	San Francisco	2003	1911 Ford Town Car	131
1	San Francisco	2003	1917 Grand Touring Sedan	128
1	San Francisco	2003	1996 Moto Guzzi 1100i	125
1	San Francisco	2003	2003 Harley-Davidson Eagle Drag Bike	124
1	San Francisco	2003	1917 Maxwell Touring Car	114
1	San Francisco	2003	2002 Yamaha YZR M1	113
1	San Francisco	2003	1932 Alfa Romeo 8C2300 Spider Sport	111
1	San Francisco	2003	1960 BSA Gold Star DBD34	101
1	San Francisco	2003	1982 Ducati 996 R	99
1	San Francisco	2003	1968 Dodge Charger	94
1	San Francisco	2003	1928 Mercedes-Benz SSK	92
1	San Francisco	2003	2002 Chevy Corvette	92
1	San Francisco	2003	1928 British Royal Navy Airplane	89
1	San Francisco	2003	F/A 18 Hornet 1/72	86
1	San Francisco	2003	1948 Porsche 356-A Roadster	86

Grafik:



Dengan analisis ini, manajemen bisa menentukan strategi bisnis, seperti promosi untuk produk dengan penjualan rendah atau meningkatkan stok untuk produk yang banyak diminati.